

Forecasting Financial Trends and Measuring Recurrence under Competing Trend Models

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Abstract

This applied study compares several established ways to define and forecast a financial trend under a common chronological validation and untouched-test protocol. Candidate methods include forecast-optimal penalized trends, classical smoothness selection, likelihood/state-space trends, autoregressive/ARIMA benchmarks, residual autoregression, and a no-change baseline. For every method, the trend forecast path is frozen at the forecast origin. We then study first-passage and recurrence behavior of future prices relative to those ex-ante paths. The central empirical question is whether forecast performance and measured recurrence are robust to the definition of trend. The numerical smoothness-selection algorithm used by one candidate is developed separately and is not a contribution of this paper.

1 Competing definitions of forecast trend

At forecast origin T , method m uses only information available through T to construct

$$\hat{\tau}_{T+k|T}^{(m)}, \quad k \geq 0.$$

Candidate method families are intentionally limited to distinct modeling principles rather than a large forecasting model zoo.

2 Chronological validation and test

Development/validation data are used for all model-specific choices. Once the protocol and model configuration are frozen, forecasts are evaluated on an untouched test block. Test outcomes do not feed back into model selection.

3 Recurrence relative to a frozen trend path

For log price x_t , define

$$g_{T,k}^{(m)} = x_{T+k} - \hat{\tau}_{T+k|T}^{(m)}.$$

A crossing time is

$$H_T^{(m)} = \inf \left\{ k \geq 1 : g_{T,k}^{(m)} g_{T,0}^{(m)} \leq 0 \right\}.$$

Additional outcomes may include tolerance-band entry, probability of recurrence by horizon, restricted-mean recurrence time, and survival/hazard summaries under right censoring.

4 Research questions

The study asks which trend definitions forecast well on untouched data, whether the preferred method changes with horizon, whether recurrence statistics are robust to the trend definition, and how recurrence depends on initial standardized distance from trend.

5 Scope

The paper is empirical/applied. It does not develop the numerical stationary-point search for normalized smoothness and does not study adaptive time-varying joint selection of (d, L, S) . Recurrence to a forecast trend is not automatically interpreted as stationary mean reversion or trading profitability.

References